

# Computational Creation and Morphing of Multilevel Rhythms by Control of Evenness

**Abstract:** We present an algorithm, instantiated in a freeware application called MeanTimes, that permits the parameterized production and transformation of a hierarchy of well-formed rhythms. Each “higher” rhythmic level fills in the gaps of all “lower” levels, and up to six such levels can be simultaneously sounded. MeanTimes has a slider enabling continuous variation of the ratios of the intervals between the beats (onsets) of the lowest level. This consequently changes—in a straightforward manner—the evenness of this level; it also changes—in a more complex, but still highly patterned manner—the evennesses of all higher levels. This specific parameter, and others used in MeanTimes, are novel: We describe their mathematical formulation, demonstrate their utility for generating rhythms, and show how they differ from those typically used for pitch-based scales. Some of the compositional possibilities continue the tradition of Cowell and Nancarrow, proceeding further into metahuman performance, and have perceptual and cognitive implications that deserve further attention.

For most of human history, musical rhythm—as with all forms of musical performance—has been a fundamentally embodied enterprise, close to dance: for example, limbs making beats by striking objects together. In all periods, however, the music involving temporal repetitions evoking dance has coexisted with “unmeasured” music, largely devoid of such impulse, such as medieval plainsong, or the unmeasured music of Couperin. Since the advent of electronic and computer technology, many creators of classical, jazz, and popular music have explored both human and artificial (nonhuman, often computer-aided) generation of rhythms, for performance and composition. Notable among these explorations in Western classical music are (1) the evasion of rhythmic regularity in some works of Iannis Xenakis and Karlheinz Stockhausen; (2) the development of serialized rhythmic structures, often more readily realized by computational than human performance; (3) irrational rhythms (such as those in instrumental composition by Brian Ferneyhough and the new complexity composers, some of which parallel those of the roughly contemporaneous free jazz and free improvisation, from John Coltrane and Cecil Taylor to Derek Bailey and Evan Parker); and (4) the establishment of noise music and its relatives, in which the surface

of the music betrays few discrete (and hence few rhythmic) features. In the popular music tradition, analog and digital (computational) sequencers, loop sequencers (e.g., Acid), and step sequencers have often been used to produce many of the characteristic sounds of pop. These sounds include the disco beats of the 1970s, “synth groups” of the 1980s, house music and hip-hop in the 1990s, and the “glitchy” beats of more recent electronic dance music.

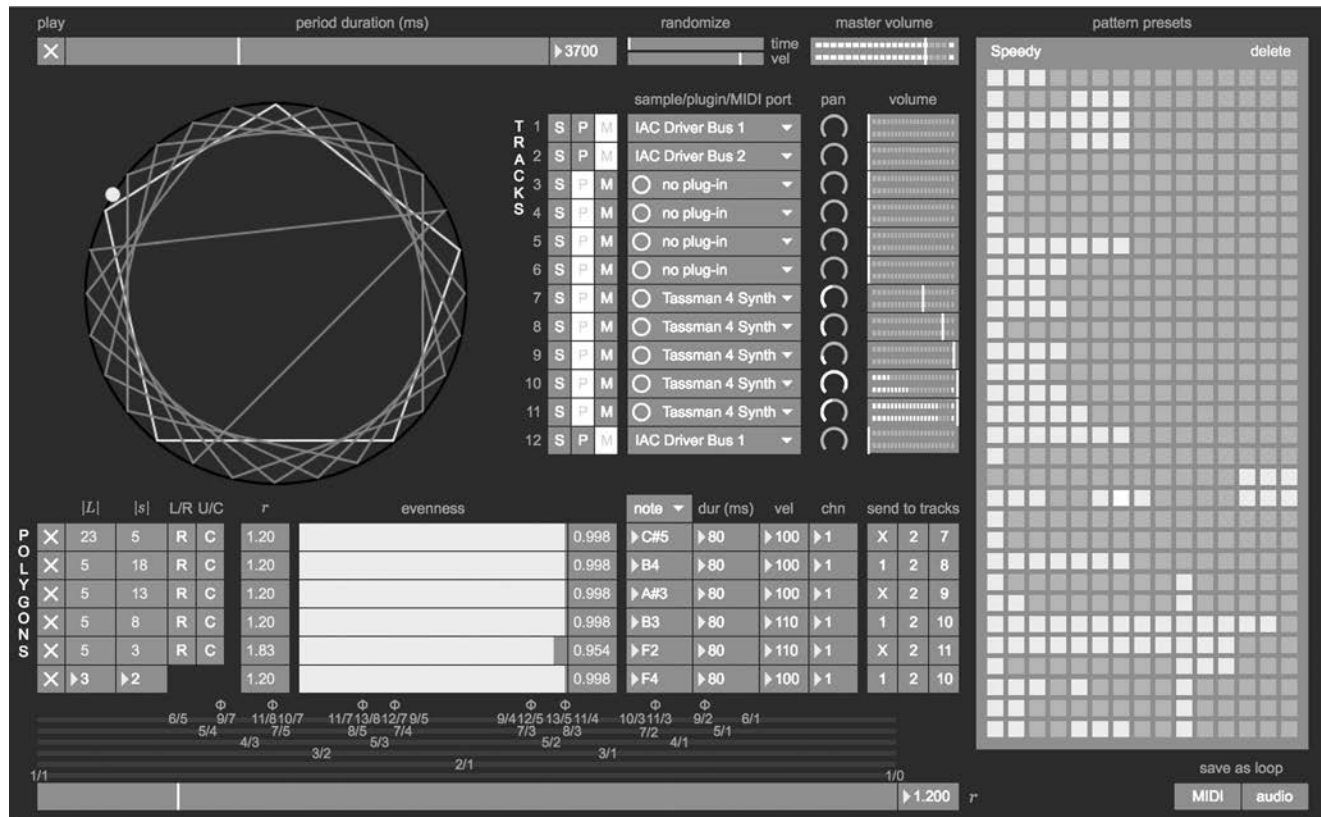
Several current computer music platforms (e.g., Max and Gibber) make it convenient to generate and control freely definable sequences of events and the number of times they are exactly repeated. In contrast to Max and Gibber, which encourage pulse flexibility and variation, many pattern-based languages (e.g., the live-coding languages *ixi lang* and *Tidal*) are primarily based on the idea of fixed isochrony of pulses, and commonly operate on bars as unitary objects. (*Isochrony* means evenly spaced in time—all inter-onset intervals are identical.) There are also numerous computational “rhythm machines” in use, published and unpublished (e.g., Dean 2003; Sioros et al. 2014). We have not, however, found a rhythm generator like MeanTimes (illustrated in Figure 1), which is capable of producing (1) a nested hierarchy of rhythmic streams that can be simultaneously systematically morphed between isochrony and nonisochrony, and (2) differing levels of complexity.

Figure 1. The user interface of MeanTimes. The hierarchy of well-formed rhythms is depicted by polygons inscribed in a circle. The large horizontal slider at the bottom is used to control beat-size

ratio. Above that are useful preset values for the slider, then settings for each of six rhythmic levels in the well-formed hierarchy, including their pitch and velocity values. At the top is a slider to

control the length of the period of rhythmic repetition. Immediately to the right of the polygons are tracks, which can host plug-ins to sonify the polygons or send MIDI to external synthesizers. To

the far right are presets, which can be saved and allow sequences of patterns to be arranged into rows or columns.



## Background

In this section, we outline the main theoretical concepts used to develop the algorithm (which is described in the subsequent section). These concepts are drawn from both musical rhythm theory and scale theory, and are applicable to the analysis of rhythm and meter in the symbolic realm, and to its physical realization (though, for the most part, we gloss over the fluctuations in tempo that typically occur in real-world performance). We also discuss some of their limitations when applied to certain familiar rhythms and suggest some possible solutions.

## Rhythm and Meter

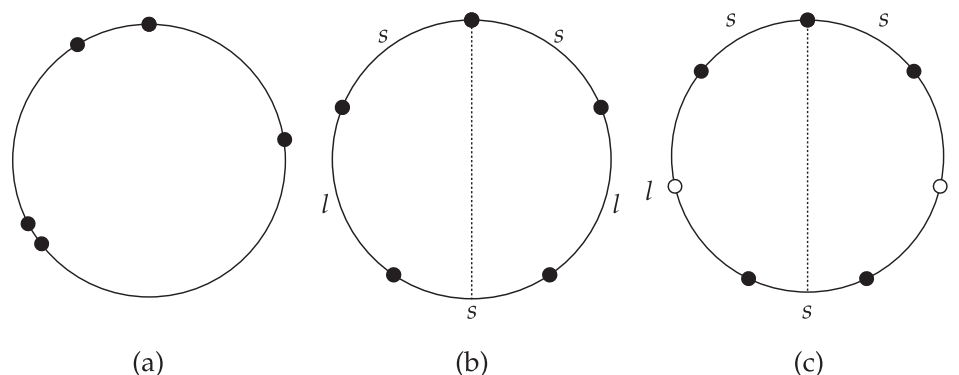
Given that our purpose is to discuss computational production and morphing of rhythm, we need to delineate the concepts of rhythm and meter that we

will use, and place them in context. We characterize a *rhythm* as a sequence of sonic *events* arranged in time, and thus primarily characterized by their inter-onset intervals. Often a rhythm comprises a number of different *streams* or *levels*. Each different stream may be played by a different instrument or by a different drum in a drum kit—for example, one stream may be played by the kick drum, another by the snare, another by the hi-hat. The decision as to which events are assigned to which stream or, indeed, whether all events are considered to belong to a single stream, depends on how much detail the analysis requires. We use the term *level* when we wish to emphasize the hierarchical nature of the streams whereby *higher levels*, which typically have higher pitches or brighter timbres, move faster than *lower levels*, which typically have lower pitches or darker timbres. When a higher-level (i.e., faster) rhythmic stream *nests* (or is a superset of) a lower-level (i.e., slower) stream, we refer to events in the

Figure 2. Some different types of event patterning that may lead to metricality: periodic (a), as is inherent in the circular representation, but there is no obvious pattern within the period; a five-beat

rhythm (b) with only two event sizes (shown as  $\ell$  and  $s$ ), repeating subsequences (e.g., there are two instances of the sequence  $s\ell$  within the period), and reflectional symmetry about the vertical axis; the

third (c) has properties similar to (b) except it is also nested by seven isochronous pulses (the open circles represent the unsounded pulses).



former as *pulses*, and events in the latter as *beats*. Under this definition, every beat must coincide with a pulse, but not every pulse must coincide with a beat (because the beat rhythm is a subset of the pulse rhythm). By extension, it is possible to have multiple rhythmic levels where each level nests all lower levels. In such a situation, we refer to events in the lowest rhythmic level under consideration as beats, and events in all higher levels as pulses. This hierarchical approach is related to that taken by Ler Dahl and Jackendoff (1983), but our terminology differs because they term all levels “beats.” Of course, there are slower metrical levels below beats, such as measures, or phrases that span several measures, or even longer-term structures, but these are not our concern here. Henceforth, we use the terms *event size* or *event length* to specify the inter-onset interval between that event and the next event on the same level. An event’s *duration* is used to specify how long that event is actually played for (which may be less, the same as, or more than its size).

In this article, our focus will be on rhythms that have a nontrivial degree of *metricality*. We take a relatively nonrestrictive functional approach to the definition of metricality: We use it to mean the extent to which a rhythm has a repeating or repeatable pattern and is perceived to have a predictable structure. Indeed, we consider a *meter* to be a mental template or dynamic process that is sufficiently stable to assign probabilities to events’ onset times. In other words, a meter is a mental representation, not a physical, sounded manifestation (which is a rhythm). This implies

that a rhythm with high metricality or, indeed a meter itself, must have some level of organization, patterning, or regularity across time. Our focus here will be on temporal patterns of the onsets of discrete events, and we will mostly ignore variations in their intensity, spectrum, or individual duration. Similar arguments will apply across all these domains, however, and they may interact to determine the resulting meter (e.g., through the influence of dynamic accents).

For events to be *patterned* we have only a general requirement that they are arranged with some regularity or symmetry that distinguishes them from a purely random arrangement. We will, however, make some concrete suggestions about possibly relevant regularities. Perhaps the most basic form of regularity is repetition over a *period*. Indeed, periodicity may be reasonably considered a prerequisite for metricality, although in much music the temporal period of a given meter may vary detectably across a piece. As illustrated in Figure 2, additional patterning may result from irregular repetitions of event sizes, or irregular repetitions of subsequences of event sizes. There may be reflective symmetries where all, or a part of, a rhythm is repeated but in retrograde form. The rhythm may exhibit some of the organizational properties identified by Toussaint (2013), or other principles such as balance (Milne et al. 2015).

These organizational principles may hold at any single rhythmic level or any mixture of levels. They may also hold for unplayed but mentally induced metrical pulses. For example, suppose a played beat rhythm that is somewhat unpatterned is a subset of

an unplayed pulse rhythm that is highly patterned. It is feasible that this beat might then induce metrical pulses analogous to this unplayed, but highly organized, pulse rhythm. For example, a beat whose events all fall at some small integer subdivision of a period may induce analogous isochronous pulses, whereas a beat whose events do not fall at such locations may not. This is concretely illustrated in Figure 2. Note how Figure 2c is a subset of a low-cardinality pulse (seven per period), while the similarly structured beat in Figure 2b is not a subset of a low-cardinality pulse (indeed, for the precise rhythm depicted, the smallest isochronous superset has 37 pulses per period, which is likely to be too fast to be readily induced as a metrical level). For this reason, we might conjecture that Figure 2c may be heard as more metrical than Figure 2b, even if the pulse level is not physically played.

Evenness (Clough and Douthett 1991; Amiot 2007) is a notable organizational principle and one that is highly pertinent to MeanTimes. The evenness of a rhythmic stream is the extent to which its events' sizes are equal (or, equivalently, the extent to which that rhythm is isochronous). When a rhythmic stream is perfectly even (i.e., isochronous), it has translational symmetry at the most granular level possible. It also has reflectional symmetry and is perfectly balanced (the mean position of the circularly arranged rhythmic events is at the center of the circle; see Milne et al. 2015), hence it has a powerful claim for high metricality. Indeed, the most common meters in Western music (e.g.,  $\frac{4}{4}$ ,  $\frac{3}{4}$ , and  $\frac{6}{8}$ ) exemplify perfect evenness at all levels. In some jazz, progressive rock, sub-Saharan African music (Rahn 1986), and Eastern European aksak (Fracile 2003), rhythms and meters typically comprise nonisochronous beats, underneath a faster isochronous pulse (which may be played or implied).

There is a class of patterns—called *well-formed* (Carey and Clampitt 1989) or *moments of symmetry* (Wilson 1975)—that can have varying degrees of evenness including perfect evenness. These patterns also have relatively numerous instances of transpositional symmetry of subsequences, and have reflectional symmetry (Figures 2b and 2c are both well-formed rhythms). Furthermore, they group into a natural hierarchy where every well-formed pattern

is nested by another. Well-formed patterns form the basis of the rhythms produced by MeanTimes. We discuss well-formedness in the section “Maximizing Evenness under Well-Formedness Constraints,” but prior to that we will provide a formal definition of evenness.

## Evenness

There are a number of different models of a pattern's evenness (Amiot 2007). We use the calculation introduced by Amiot (2009), for the following reasons: It makes no assumption of an underlying isochronous pulse, it has a clear interpretation in terms of conventional circular statistics (Fisher 1993), and it can be relatively easily understood. We provide a formal definition in the following but, for those wishing to skip the mathematics, a simple summary is that evenness is the similarity of a rhythm with  $M$  events to an isochronous (hence perfectly even) rhythm also with  $M$  events.

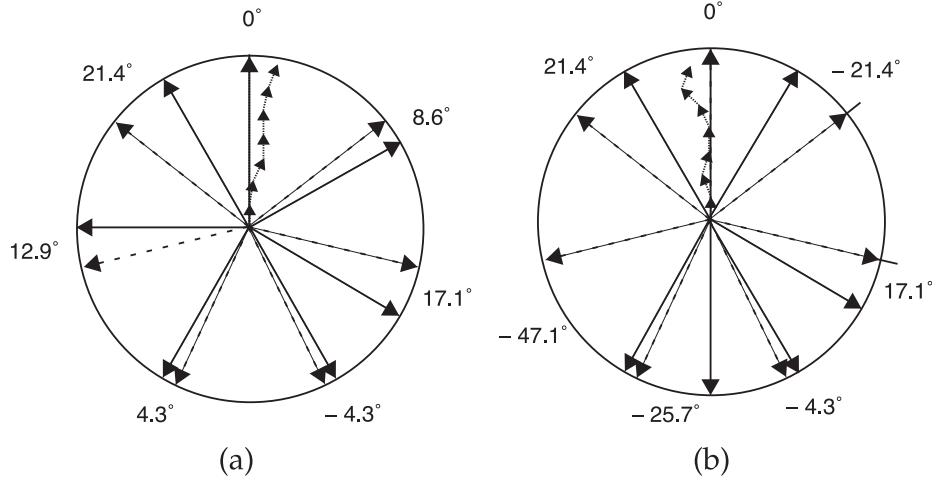
First, the  $M$  onset times are placed, in time order, into a vector. These times are then normalized by subtracting the lowest time value from all of them and then dividing them by the length of the whole period so the resulting values start from 0 and are all less than 1. This vector is denoted  $\mathbf{x}$  and its  $M$  elements are indexed by  $m = 0, 1, 2, \dots, M - 1$ . The elements of  $\mathbf{x}$  are then mapped to the unit circle in the complex domain by  $\mathbf{z}[m] = e^{2\pi i \mathbf{x}[m]}$ . This results in a vector of  $M$  complex numbers, where the phase (angle or argument) of the  $m$ th element represents the  $m$ th event's normalized onset time, and all events have a magnitude of 1.

Each such complex number can itself be thought of as a vector extending from the center of the circle to its perimeter; hence  $\mathbf{z}$  can be thought of as a vector of vectors. For example, let us consider two rhythms all of whose beats fall at a twelfth equal division of the period. The first is analogous to an ordered diatonic pitch-class set and can be represented by  $\mathbf{x} = \{0/12, 2/12, 4/12, 5/12, 7/12, 9/12, 11/12\}$ ; the second comprises two “clusters” of beats  $\mathbf{x} = \{0/12, 1/12, 4/12, 5/12, 6/12, 7/12, 11/12\}$ . The mapping of each of these two vectors to the complex unit circle is illustrated by the solid vectors (arrows)

Figure 3. A graphical representation of the evenness calculation described in the main text. The solid-line vectors show the “diatonic” rhythm (a) and the “cluster” rhythm (b), both

having seven events. The dashed-line vectors show the seven isochronous locations to which the rhythms are compared. The numbers around the circle are the angles between the  $m$ th solid-line

rhythm vector and the  $m$ th dotted-line isochronous vector—the squiggly line is the sum of seven vectors having these respective angles, each with a normalized length of  $1/7$ .



in Figure 3 (in this visualization, the imaginary unit  $i$  is at 3 o'clock, the real unit  $1$  is at 12 o'clock, and angles are measured clockwise from the latter—this is a diagonal reflection of the conventional mathematical visualization, but results in the familiar clockface type representation usually depicted in rhythm-related research). The first element of  $\mathbf{x}$ , which is 0 in both of these rhythms, is mapped to 1 in the complex plane and so goes to the 12 o'clock position, while successive elements proceed clockwise around the circle (the angles written around the circle will be explained forthwith).

The next step is to take the discrete Fourier transform of this complex vector, and the magnitude of the first coefficient gives the evenness of the pattern:

$$\begin{aligned} \text{evenness} = |\mathcal{F}\mathbf{x}[1]| &= \left| \frac{1}{M} \sum_{m=1}^M \mathbf{z}[m] e^{-2\pi i m/M} \right| \\ &= \left| \frac{1}{M} \sum_{m=1}^M e^{2\pi i (\mathbf{x}[m] - m/M)} \right|. \quad (1) \end{aligned}$$

The summand in the second line,  $e^{2\pi i (\mathbf{x}[m] - m/M)}$ , makes it clear that what is being calculated, for each value of  $m$ , is a unit-length vector whose angle is equivalent to the directional (angular) displacement between the  $m$ th event of the beat structure under investigation and the  $m$ th event of an isochronous

meter (both containing  $M$  events). In Figure 3, the seven isochronous locations are shown with dashed vectors, and the clockwise displacement from the  $m$ th isochronous location to the  $m$ th beat is shown by the respective number outside the circle.

The magnitude of the first coefficient,  $|\mathcal{F}\mathbf{x}[1]|$ , is the length of the *resultant* vector, which is the sum of the  $M$  previously mentioned unit-length vectors whose angles correspond to the respective displacements. Clearly, the smaller the variance of these displacements, the longer the resultant vector will be. When the events are maximally even—which implies all their displacements are identical, so all  $M$  summands are parallel vectors—the normalization by  $1/M$  ensures that the length of the resultant (i.e., the evenness value) is unity. In Figure 3, this summation of the angular displacement vectors is depicted by the seven small vectors extending from the center of the circle outwards. Each vector has a length of  $1/7$  because of the normalization. The innermost vector has an angle of  $0^\circ$  (vertical) because the displacement between the first beat and the first isochronous location is zero; the next vector outwards has an angle corresponding to the displacement between the second beat and the second isochronous location ( $8.6^\circ$  in Figure 3a,  $-21.4^\circ$  in Figure 3b); and so on. Note how the vector resulting from the sum of these displacement vectors is slightly longer in Figure 3a (its length is 0.988) than it is in Figure 3b (its length is 0.924); hence



the diatonic rhythm is more even than the cluster rhythm.

Using the standard definition of circular variance (Mardia 1972; Fisher 1993), metrical evenness is equivalent to unity minus the circular variance of the displacements. Importantly, the calculation can be applied equally to any single rhythmic stream (e.g., beat or pulse) or any combination of such streams. In all cases, the  $M$  rhythmic events are compared with  $M$  isochronous events occupying the same period.

#### *Maximizing Evenness under Metrical Constraints*

Clearly, this definition implies that a maximally even rhythm of  $M$  beats has isochronic beats (a simple result: this is *perfect evenness*). As soon as certain common constraints are applied to the metrical structure, however, isochronous beats that also coincide with isochronous pulses become impossible to achieve, and the resulting patterns are more interesting. The most obvious example is when there are  $M$  beats in the same period as  $N$  isochronous pulses, and  $M$  and  $N$  are *coprime* (this means that their greatest common divisor is unity, which can be concisely written as  $\gcd(M, N) = 1$ ). In this case, there is no way for the beats to be isochronous and also to coincide with pulses (the latter being a requirement under our previously given definitions of beat and pulse).

Consider a twelve-pulse, five-beat rhythm: the two numbers are coprime so, as just indicated, there is no way for the beats to fall on pulses and also be isochronous (perfectly even) under the constraint that all beats must coincide with a pulse. It is possible to maximize the evenness of the beats under these same constraints, however: We have to choose the most even arrangement of five beats (i.e., the beats are not isochronous), whose total length equals twelve isochronous pulses. Clearly, there are numerous such patterns, but one of them will be maximally even. To be more concrete, let there be two beats of size  $\text{♩}$  (dotted quarter-note) and three beats of size  $\text{♩}$  (quarter note). Ignoring *rotation* (i.e., the position of the period commencement or the “bar line” relative to the events), there are just

two ways these beats can be arranged ( $\text{♩} \text{♩} \text{♩} \text{♩} \text{♩}$  or  $\text{♩} \text{♩} \text{♩} \text{♩} \text{♩}$ ). Of the two possibilities, the latter pattern is closer to the isochronous beat pattern (the displacements have less variance) and is, hence, the pattern of greatest evenness under these constraints. Indeed, this latter pattern is maximally even with respect to all possible patterns of five beats in a twelve-isochronous-pulse period (i.e., including those with more than two beat sizes). Generally, for any  $\gcd(M, N) = 1$ , the maximally even pattern will have no more than two step sizes. This method for patterning events corresponds precisely to Justin London’s (2004) well-formed constraint WFC 6 (version 1).

#### *Maximizing Evenness under Well-Formedness Constraints*

Setting  $\gcd(M, N) = 1$  is not, however, the only useful constraint under which to maximize evenness. There is an alternative constraint that is particularly attractive because it makes no assumption about a superset of isochronous pulses. As noted earlier, the resulting patterns are known as “well-formed” (Carey and Clampitt 1989) or, in a tuning-theory context, “moments of symmetry” (Wilson 1975). (As shown by these citations, well-formedness was first applied to pitch-based scales. It is natural to extend this property from scales, which are events distributed in pitch with repetition at the octave, to rhythms, which are events distributed in time with repetition at the period.) Well-formed patterns are a superset of the previously described  $\gcd(M, N) = 1$  patterns, and they are a method for generalizing the latter into contexts with a nonisochronous pulse or no underlying pulse.

The *well-formed constraint* is that there are  $j$  beats of size  $\ell$ , there are  $k$  beats of size  $s$ , where  $j$  and  $k$  are coprime, and there are no beats with a size that is not  $\ell$  or  $s$ . For a period of total length  $d$ , this implies  $d = j\ell + ks$ . In a general setting,  $\ell$  and  $s$  can take any possible sizes. For this article, however, we will apply the additional constraint that  $s \leq \ell$ ; that is, that the  $\ell$ -sized beats are “long” or “large” and the  $s$ -sized beats are “short” or “small.” (In a pitch-based context, the use of  $\ell$ , or  $L$ , and  $s$  follows

the notational convention introduced by Erv Wilson, and one that is followed by much of the microtonal community; e.g., [xenharmonicwiki.com](http://xenharmonicwiki.com).) Most of the properties we outline subsequently also hold for  $s > \ell$ , but the  $s \leq \ell$  constraint simplifies some of the exposition.

We additionally denote the step-size ratio  $r = \ell/s \in [1, \infty)$ , after Blackwood's (1985) use of  $R$  to describe the ratio between the sizes of major and minor seconds in different tunings of the diatonic scale. This ratio is one of the principal parameters used in MeanTimes to control the rhythmic output (see Figure 1). It is useful because, when  $r$  is a rational number whose *reduced fraction* is  $a/b$  (i.e., a fraction where the numerator and denominator are both divided by their greatest common divisor), every beat falls at an  $N$ th division of the period, where  $N = ja + kb$ , where  $j$  is the number of long beats, and  $k$  is the number of short beats. Therefore, if  $N$  is sufficiently small, the meter has a perceptibly isochronous pulse rhythm. In MeanTimes, these faster pulse rhythms can be sounded, if required, along with the main beat. Conversely, if  $N$  is sufficiently large, or  $r$  is irrational and sufficiently distant from the nearest low-integer ratio, it is more reasonable to consider the pulse to be nonisochronous. Furthermore, when  $r < 2$ , the scale is *proper* (Rothenberg 1978) or, equivalently, *coherent* (Balzano 1982), which means the total length of any  $m$  consecutive beats is always larger than the total length of any  $m - 1$  consecutive beats.

Justin London (2004), for example, has suggested that incoherences (where the above property does not hold) may make the resulting meters less stable or, when the clumping of beat sizes is extreme, they may be heard as alternating meters of different lengths. For example, the non-well-formed  $3 + 3 + 2 + 2 + 2$  (a common compositional pattern, used in Bernstein's "America" and often found in flamenco) may be heard as a repetitive alternation of  $3 + 3$  and  $2 + 2 + 2$ . In this rhythm, the first two consecutive beats—each of size 3—have a total length of 6, and the last three consecutive beats—each of size 2—also have a total length of 6; hence there is an incoherence.

Interestingly, the maximally even arrangement of the two beat sizes  $\ell$  and  $s$  is invariant across

all possible ratios  $1 \leq r \leq \infty$ . For example,  $4 + 3 + 4 + 3 + 3$  (so  $r = 4/3$ ),  $3 + 2 + 3 + 2 + 2$  (so  $r = 3/2$ ), and  $3 + 1 + 3 + 1 + 1$  (so  $r = 3/1$ ) are all maximally even arrangements of the beat sizes they comprise because they all follow the same abstract "well-formed pattern" ( $\ell s \ell s s$ ). Such patterns minimize, as much as possible, the clumping together of the  $\ell$ -sized beats and the clumping together of the  $s$ -sized beats—put differently, the  $\ell$ -sized beats are spaced as far apart as possible, and so are the  $s$ -sized beats.

By covarying the values of  $\ell$  and  $s$  (and hence  $r$ ), while simultaneously ensuring that the length of the period ( $j\ell + ks$ ) is constant, it is possible to move through a continuum of metrical structures, passing through and connecting meters with differing numbers of isochronous pulses. It is worth noting that there are two instances along this continuum that are *degenerate*: when  $\ell = s$  (so  $r = 1$ ), and  $s = 0$  (so  $r \rightarrow \infty$ ). At these values, the resulting beats are perfectly isochronous with, respectively,  $j + k$  and  $j$  beats occurring in each period. This means that smoothly changing the  $r$  value over 1 to  $\infty$  smoothly interpolates between these two isochronous beats, while keeping the period's length constant: the  $j + k$  events are reduced down to  $j$  by having pairs of events occurring simultaneously.

Other than maximizing evenness (given the above constraints), well-formed patterns (both meters and scales) have several other consequent mathematical properties, which we will elaborate elsewhere (these properties are also discussed in more depth in Milne et al. 2011). It is worth noting at this stage, however, that well-formed patterns have reflectional symmetry: When the total number of beats is odd, the axis of reflection falls on one of the beats; when the number of beats is even, the axis of reflection falls mid-way between two beats (and may fall between pulses too, depending on the value of  $r$ ). In both cases their  $\ell, s$  alphabetic representation is, therefore, a circular palindrome, as illustrated in Figures 2b and 2c.

Because well-formed rhythms comprise no more than two different beat lengths, it follows that they will have a lower beat-length entropy in comparison with the huge number of possible rhythms containing three or more beat lengths. As

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discussed later, there are other non-well-formed patterns that also contain just two beat sizes having similarly low entropy.

Another highly pertinent property of a well-formed pattern is that it is always nested inside a larger well-formed pattern. By *nested*, we mean that every event in one pattern (the subset) coincides with an event in the other (the superset). This can be most easily exemplified by musical scales, where scale-step sizes are treated as analogous to beat sizes (early elaborations of the analogy between a meter and a pitch scale were offered by Rahn 1975; Mazzola 1990; Monahan 1993; Cowell 1996 [1930]). For example, in a pitch-based context, the well-formed perfect fifth D–A ( $\ell s$ ) is nested in the well-formed “tetractys” D–G–A ( $\ell s \ell$ ), which is nested in the well-formed five-note pentatonic scale D–F–G–A–C ( $\ell s s \ell s$ ), which is nested inside the well-formed seven-note diatonic scale D–E–F–G–A–B, C ( $\ell s \ell \ell \ell s \ell$ ), which is nested inside the well-formed twelve-note chromatic scale D–D $\sharp$ –E–F–F $\sharp$ –G–G $\sharp$ –A–A $\sharp$ –B–C–C $\sharp$  (where, typically,  $\ell = s$ ). But this is just one example of such a hierarchy. This means that the hierarchical levels, which can play concurrently, provide a natural analogue for the hierarchical structure of beats and pulses in the common conception of meter. This also raises another possibility, which is that any beat may be displaced to a neighbouring pulse. This provides a means to modify well-formed patterns into closely related (hence still highly structured) non-well-formed patterns. Both these aspects are explored in more detail in the section “MeanTimes: The Application and its Design.”

### Creating Real-World Meters and Rhythms

Well-formed rhythms are a common strand within theories of rhythm and meter (e.g., London 2004; Toussaint 2013). And, interestingly, it has been observed that well-formed rhythms are common in sub-Saharan music (Rahn 1986), and irregular groupings of strong beats in aksak music often fall into such well-formed patterns, too (Fracile 2003). Western music also contains numerous examples of well-formed rhythms, such as those that formed

the basis for the rules and WFCs discussed by London.

Despite the appealing conceptual properties of well-formed patterns, however, there are examples of important rhythms (and scales) that are not well-formed. For example, the two beat lengths may not be distributed with maximal evenness (e.g., the  $3 + 3 + 2 + 2 + 2$  given earlier) and, in this case, the pattern may perceptually resolve into an alternation of two (well-formed) meters  $3 + 3$  and  $2 + 2 + 2$ . Alternatively, the rhythmic pattern may contain more than two differing beat lengths; for example, the five-beat, 16-pulse *son* clave pattern ( $3 + 3 + 4 + 2 + 4$ ). So this raises a question of whether there is a principled way to extend the framework of well-formed rhythms in such a way as to include a useful subset of non-well-formed rhythms.

London relaxes his constraints in the second version of WFC 6 precisely to enable such meters. As we show later, our conception of meter as hierarchies of well-formed rhythms allows such non-well-formed meters to occur naturally by substituting a beat on one level with a neighboring pulse from the next higher level. This is analogous to chromatic alteration in a diatonic-chromatic context, whereby one might flatten or sharpen a diatonic scale degree to obtain a different—yet still highly structured—scale. For example, flattening the third degree of the major scale (C–D–E–F–G–A–B) produces the ascending melodic minor scale (C–D–E $\flat$ –F–G–A–B). This latter scale comprises two step sizes, but they are now not arranged with maximal evenness, though the scale is still reasonably even (its evenness value is not far below 1). Taking a common metrical example, the *son* clave pattern  $3 + 3 + 4 + 2 + 4$  is not well-formed (after all, it has three beat lengths). But it can be simply constructed as a well-formed rhythm whose fifth event has been “flattened”; that is, it is an altered version of the well-formed  $3 + 3 + 4 + 3 + 3$ . Such displacement of the beat might also occur because of repeated accentuation of the specified higher-level pulse note.

We do not mean to suggest here that well-formedness and evenness are necessary principles of rhythm production. Indeed, there are many different

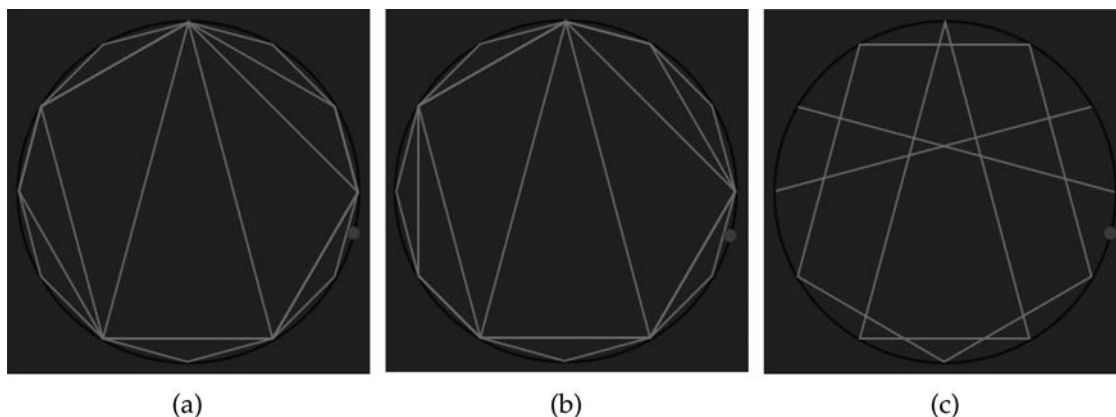


Figure 4. Three well-formed rhythms, each comprising six rhythmic levels visualized in MeanTimes as polygons inscribed in a circle. Each polygon vertex represents a beat or pulse. The small disk between 3 and 4 o'clock rotates clockwise around the

circle and represents a “playhead.” When the playhead passes over a vertex, the corresponding MIDI note is played. In (a), Level 0 is the upwards pointing isosceles triangle (its vertices correspond to the twelve-tone equal temperament [12TET] pitch

classes C, F, and G), Level 1 is the irregular pentagon (pentatonic), Level 2 is the irregular heptagon (diatonic), Level 3 is the regular dodecagon (chromatic). In (b), the diatonic Level 2 has been set to counterclockwise

subsequently). In (c), Levels 1–3 from (a) have been set to complementary mode (discussed subsequently) so Levels 1 and 2 become digons (straight lines), and Level 3 becomes an irregular pentagon (pentatonic).



strategies that may be taken—for example, using balance rather than evenness (Milne et al. 2015). We suggest, however, that they are useful principles relevant to both rhythm production and probably to perception. As such, they provide a useful means to reduce the space of rhythms into one that can be smoothly parameterized and that also contains a wide variety.

## MeanTimes: The Application and its Design

Developing from the analogy between (rhythmic) meter and (pitch) scale introduced earlier, and the consideration of evenness and well-formedness, MeanTimes is constructed as a MIDI rhythm generator providing a rhythmic parallel to the pitch-based manipulations used by the freeware Dynamic Tonality synthesizers and sequencers detailed by Milne, Sethares, and Plamondon (2008), Sethares and coworkers (2009), and Precht and colleagues (2012). MeanTimes is built in Cycling ’74’s Max, and can be downloaded from [www.dynamictonality.com](http://www.dynamictonality.com) as a standalone application for Windows and Mac OS X.

As detailed subsequently, MeanTimes provides a number of parameters to shape six separate rhythmic levels of a well-formed hierarchy. The user can set the length, in milliseconds, of the period of repetition (denoted  $d$ ), and the numbers of long beats and short beats into which this period

is divided for the lowest-level rhythm (in this article, these numbers are denoted  $j$  and  $k$ , but have different designations in MeanTimes’s interface). The lengths of these beats are denoted  $\ell$  and  $s$ , respectively: Their ratio  $r = \ell/s$  can be smoothly varied with a slider along the entire continuum that preserves that rhythm’s structure (i.e., given  $j$  and  $k$ , the range  $1 \leq r < \infty$  within which  $s \leq \ell$ ). The sizes of  $\ell$  and  $s$  are automatically calculated to ensure the period’s length is invariant across all values of  $r$ . It is also possible to selectively sound a number of “higher-level” well-formed rhythms that nest the “Level 0” beat rhythm, and that smoothly change as a function of  $r$  over its entire range. As discussed earlier, at certain values of  $r$ , some of these higher-level rhythms will comprise perfectly isochronous pulses.

As shown in close-up in Figure 4, each rhythmic level is visualized as a polygon inscribed in a circle. A “playhead,” depicted as a small disk, rotates around the circle and, whenever it “hits” a polygon vertex, a MIDI note is sent out with a pitch, duration, channel, and output port that has been designated for that polygon by the user.

Our approach to parameterizing and explaining well-formedness is novel because of our focus on the size of  $r$ , rather than the size of a generating interval, which, with the notable exception of Blackwood (1985), is the more usual way to parameterize well-formed pitch-based scales (Wilson 1975; Erlich 2006;

Milne et al. 2011). Indeed, our approach can be seen as a generalization of Blackwood's in that it can additionally encompass any possible well-formed pattern and the well-formed patterns that embed it (not just diatonic and chromatic).

In a rhythmic context, parameterizing by  $r$  is sensible because of the importance of isochronies (and deviations therefrom), which occur when  $r$  is a low integer ratio, whereas in a pitch context, the acoustic properties of intervals between nonadjacent scale members become important. For example, the interval used to generate a well-formed scale is the most prevalent in that scale, which may explain the importance of the pentatonic, diatonic, and chromatic scales because they are traditionally generated by the most acoustically simple and consonant pitch-class interval other than the octave, the perfect fifth/perfect fourth (Milne, Laney, and Sharp 2016).

### Generating and Parameterizing Well-Formed Rhythms

The use of  $r$  as our primary parameter of interest leads to the following mathematical description. To provide the necessary background, we first outline the word-theory characterization of well-formedness recently developed by Clampitt, Domínguez, and Noll (2009), and related earlier work by Wilson (1991). We then explain our novel extensions of this theory: extensions that allow us to create parameterized multileveled well-formed rhythms.

#### Well-Formed Words

In order to create a well-formed rhythm, it is necessary to know the pattern in which its long and short beats are arranged; in other words, it is necessary to know its *well-formed word* over the alphabet of the letters  $\ell$  and  $s$  (Clampitt, Domínguez, and Noll 2009). Up to rotation, each rhythm is uniquely defined by the numbers ( $j$  and  $k$ ) of long and short beats it contains and, as described earlier, the two letters (beat sizes) are arranged so they are clumped as little as possible. The well-formed word

is equivalent to the Christoffel word of slope  $j/k$ , or any *conjugate* thereof (Berstel et al. 2008; Clampitt, Domínguez, and Noll 2009). In this context, a conjugate is some rotation (also commonly called a “circular shift”) of the word, where a sequence of letters are removed from the end of the word and added to the start.

As demonstrated in principle (the details and context differ) by Erv Wilson (1991), well-formed words can be generated by the following simple algorithm. Start with the word  $\ell s$ , then make two different pairs of substitutions (each pair of substitutions is termed a *morphism*): the first morphism is  $\ell \mapsto \ell s$  and  $s \mapsto \ell$ ; the second morphism is  $\ell \mapsto \ell s$  and  $s \mapsto s$ . Applying the first morphism to  $\ell s$  gives  $\ell s \ell$ ; applying the second morphism gives  $\ell s s$ , hence two new words have been generated. The two morphisms are then re-applied to the two new words, resulting in four more new words:  $\ell s \ell \ell s$ ,  $\ell s s \ell s$ ,  $\ell s \ell \ell$ , and  $\ell s s s$ . Applying the morphisms again results in eight new words and, in general, the  $i$ th iteration provides  $2^i$  new words, making a total of  $2^{i+1} - 1$  words. Both morphisms can be understood as splitting each long beat into a new long and a new short beat ( $\ell \mapsto \ell s$ ). In the first morphism, the size of the new long beat corresponds to the size of the old short beat ( $s \mapsto \ell$ ); in the second morphism, the new short beat is the same as the old short beat ( $s \mapsto s$ ). These two morphisms produce only Christoffel words or their conjugates because they are compositions of the standard Christoffel morphisms: the first morphism is  $G \circ E$ ; the second is  $\tilde{D}$  (Berstel et al. 2008).

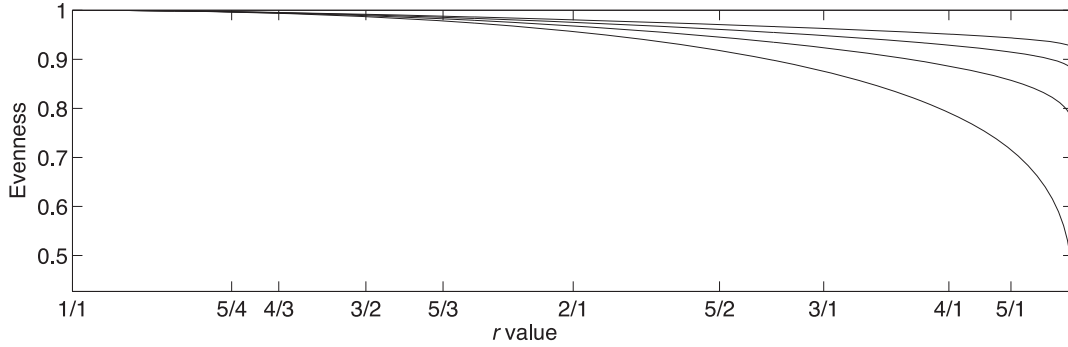
#### Calculating Beat Sizes

As described earlier,  $(j, k)$  defines the well-formed word—the ordering of the beats of sizes  $\ell$  and  $s$  in the period. It does not, however, define the values of  $\ell$  and  $s$ . For that, the length (temporal duration)  $d$  of the period and the ratio  $r$  of the two step sizes are required. As previously established,  $d = j\ell + ks$ , and by definition,  $r = \ell/s$ . These two equations imply that  $\ell = dr/(jr + k)$  and  $s = d/(jr + k)$ , which, in conjunction with the well-formed word, allow us to calculate the precise timings of all beats as a function of the period length  $d$ , the numbers of

Figure 5. The evenness of all four five-beat well-formed rhythms as a function of the ratio of long and short beat sizes. The patterns are, from highest evenness to lowest

evenness: (4ℓ, 1s), (3ℓ, 2s), (2ℓ, 3s), and (1ℓ, 4s). The minimal evenness values occur when  $r \rightarrow \infty$ . In general, for any well-formed rhythm with just one long

beat—e.g., (1ℓ, 4s)—this minimal evenness value approaches zero, for the other well-formed rhythms it approaches values greater than zero.



long and short beats ( $j, k$ ), and the beat-size ratio  $r$  (all of which can be controlled in MeanTimes's user interface). It is also trivial to generate different "modes" (starting points) for the rhythms by rotating the well-formed word.

Because  $r$  has an infinite range, we scale it by adjusting a parameter  $t \in [0, 1]$ , such that  $r = 1/(1 - t)$ . Hence  $t = 0$  implies  $r = 1$ , and  $t = 1$  implies  $r = \infty$ . This ensures that when the  $r$ -slider is in its halfway position (i.e.,  $t = 0.5$ ), the resulting  $r$  value is  $2/1$ . This is a particularly important beat-size ratio because—as shown in the next section—it always implies that the next higher-level well-formed rhythm (which may be thought of as comprising pulses, some of which coincide with the lower-level beat) is isochronous.

As previously discussed, as the value of  $r$  is moved throughout its range, it affects the evenness of the resulting rhythm. When  $r = 1$ , the long and short beats are of identical size, so the rhythm is maximally even (isochronous). As  $r$  is increased, the evenness reduces. For any given  $r$  value, however, evenness also varies across different well-formed rhythms. Usually, the evenness of well-formed rhythms, all of the same number of events  $j + k$ , is ordered by the size of  $j$ . That is, the greater the number of long beats, the greater the evenness of that well-formed rhythm at any given  $r$  value. This is illustrated in Figure 5, which shows the evenness values for all five-beat well-formed rhythms over  $1 \leq r < \infty$ . Because the minimal evenness value differs across rhythms, and also to keep the interface simple, we do not allow the user direct control of the evenness value. The changes in evenness (for each

level) are displayed in the interface of MeanTimes, and they arise from the user-controlled changes in  $r$ , or the other parameters.

### A Nested Hierarchy of Rhythmic Levels

As previously discussed, every well-formed rhythm is nested inside another higher-level well-formed rhythm. This allows a hierarchy of such rhythms to be played simultaneously or successively: MeanTimes's interface (Figure 1) includes a set of six checkboxes to toggle playback of the user-defined beat rhythm, and the next five higher levels of pulses.

Let each successive rhythmic level be denoted  $i$ , where  $i = 0$  is the beat rhythm (whose  $j$  and  $k$  values are entered by the user),  $i = 1$  is the Level 1 well-formed pulse rhythm that nests it,  $i = 2$  is the Level 2 well-formed pulse rhythm that nests the Level 1 pulse rhythm,  $i = 3$  is the Level 3 well-formed pulse rhythm that nests the Level 2 pulse rhythm, and so forth. This means that any meter is nested inside a hierarchy of higher-level pulse rhythms, with each higher-level pulse rhythm level being a superset of the previous.

We now make some more general statements about the relationships of the well-formed rhythms of successive levels. First, let us consider the numbers of long and short pulses in level  $i + 1$  as a function of level  $i$ :

$$(j_{i+1}, k_{i+1}) = \begin{cases} (j_i + k_i, j_i), & r_i \leq 2/1, \\ (j_i, j_i + k_i), & r_i \geq 2/1. \end{cases} \quad (2)$$

Second, let us consider the sizes of the long and short pulses:

$$(\ell_{i+1}, s_{i+1}) = \begin{cases} (s_i, l_i - s_i), & r_i \leq 2/1, \\ (l_i - s_i, s_i), & r_i \geq 2/1. \end{cases} \quad (3)$$

Equations 2 and 3 are a direct consequence of the  $(i + 1)$ th level being derived from the  $i$ th level by the application of one of the morphisms described at the start of this section ( $\ell \mapsto \ell s$  and  $s \mapsto \ell$ , or  $\ell \mapsto \ell s$  and  $s \mapsto s$ ).

Given that  $r_{i+1} = (\ell_{i+1})/(s_{i+1})$ , and substituting  $\ell_i = r_i s_i$  into Equation 3 implies

$$r_{i+1} = \begin{cases} \frac{1}{r_i - 1}, & r_i \leq 2/1, \\ r_i - 1, & r_i \geq 2/1. \end{cases} \quad (4)$$

Some pertinent corollaries flow from these equations. Usually, once  $r_0$  is set,  $r_1, r_2, r_3$ , and  $r_4$  will each take different values. We will point out some interesting cases. First, Equation 4 shows that  $r_0 = 2/1$  implies  $r_1 = 1/1$ . Furthermore, Equation 2 shows that when  $r_0 = 2/1$ , there are two possible solutions for  $(j_1, k_1)$ , which are  $(j_0 + k_0, j_0)$  and  $(j_0, j_0 + k_0)$ . Together, these imply that both Level 1 rhythms are isochronous with  $2j_0 + k_0$  pulses, so the transition between them, which occurs when  $r_0 = 2/1$ , is *seamless* (no discontinuity).

Second, again when  $r_0 = 2/1$ , the rhythm two levels higher has  $r_2 \rightarrow \infty$ . Furthermore, Equation 2 shows that this rhythm has two possible forms, which are  $(2j_0 + k_0, j_0 + k_0)$  and  $(2j_0 + k_0, j_0)$ . Together, these imply that both are isochronous rhythms with  $2j_0 + k_0$  pulses to the period because the small-sized steps have shrunk to zero size. This means the transition across this boundary is also seamless. By iterating this procedure, it is trivial to show that the same isochronous  $2j_0 + k_0$  pulse rhythm is the boundary between all pairs of higher-level well-formed rhythms above  $r_0 = 2/1$ .

These two corollaries mean that any given pulse-pattern level has no discontinuities across the full range of legal  $r_0$  values. Hence it makes sense to allow users to enable or disable the playing of higher-level well-formed pulse-patterns for each level  $i$  rather than, for example, allowing them to

choose from among a variety of higher-level pulse-patterns as and when they become available across the  $r_0$  continuum. In this way, multiple rhythmic levels can play simultaneously, and the transitions across all well-formed boundaries as the  $r_0$ -slider's value is adjusted will always be seamless.

A final corollary is that whenever  $r_0$  is irrational, there will never be a higher rhythmic level that is perfectly even (there is no  $r_i$  that equals 1). For many irrational numbers, however, the  $r_i$  for certain levels may closely approximate 1. Interestingly, there is a class of irrational  $r_0$  values where  $r_i$  values never come close to 1, and produce what we call *deeply nonisochronous* rhythms. These are  $r_0$  values of the form  $(a\phi + c)/(b\phi + d)$ , where the fractions  $a/b$  and  $c/d$  are adjacent members from level  $s$  and level  $s + 1$  of the Stern-Brocot tree,  $a/b < c/d$ , and  $\phi = (1 + \sqrt{5})/2 \approx 1.618$  is the golden section. When  $r_0$  takes such a value, all  $r_{i \geq s} = \phi$  (Wilson 1997). As can be seen by substituting  $\phi$  into Equation 4, whenever a level has  $r = \phi$ , all higher levels get “stuck” with this value and so can never come close to 1 or  $\infty$  (the only two values that imply isochrony). The only other value that gets stuck in this way is  $\infty$  (which implies isochrony). Indeed, all rational  $r_0$  values have higher levels that ultimately converge on  $\infty$ , while almost all irrational  $r_0$  values have higher levels that either “wander” through different  $r$  values (in all likelihood some of which will be close to 1 or  $\infty$  and hence close to isochrony) or converge to  $\phi$ . We denote the  $r_0$  values that reach  $\phi$  on the  $s$ th level  $\phi_s$ . These values are shown above the  $r$ -slider in MeanTimes. Other values of  $r_0$  may also produce rhythms with no perceptible isochrony because the (approximately) isochronous level is too fast, but the  $r_0 = \phi_s$  values minimize convergence to isochrony in the mathematical limit.

### An Example of a Nested Hierarchy

The structure of well-formed rhythms across different levels and  $r_0$  values is therefore complex, yet highly organized and patterned. We now give a more concrete illustration of this structure, and of the mathematics introduced in the previous subsection, by focusing on the  $(2\ell, 3s)$  well-formed beat rhythm and the hierarchy of pulse rhythms that nest it over

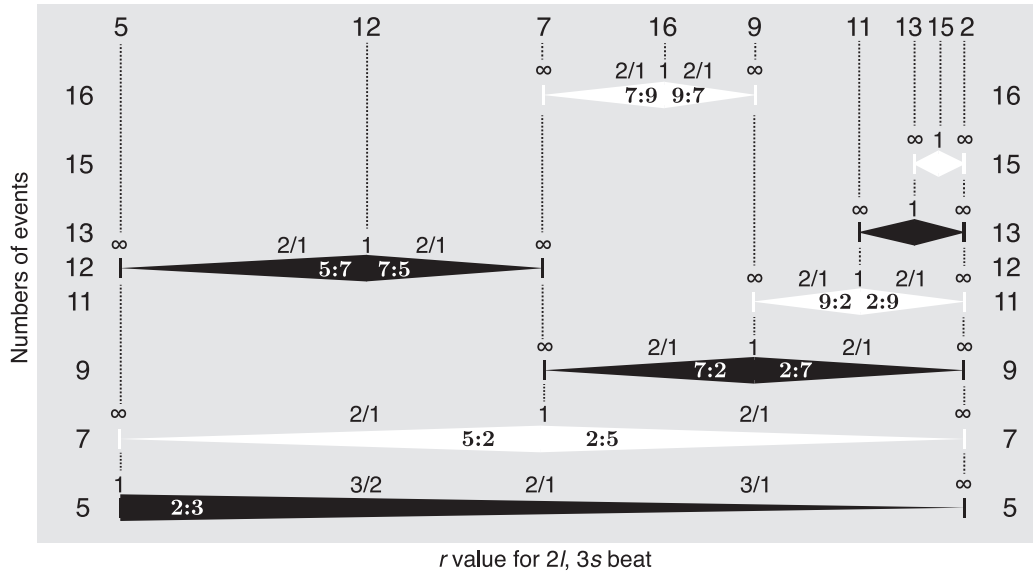


Figure 6. The hierarchy of well-formed rhythms nesting a five-beat (2ℓ, 3s) rhythm across the latter's full range of  $r_0$  values. The bottom half-diamond represents the beat rhythm, the diamonds

above represent the nesting pulse rhythms. The horizontal extent of each rhythm's diamond shows the  $r_0$  values over which it exists. The numbers down the side show the number of events in the rhythms,

while the numbers inside each half-diamond (e.g., 5:2) show their numbers of long and short pulses. Some  $r_i$  values for each well-formed rhythm are shown directly above its diamond. The dotted

vertical lines show the locations where isochronic pulses occur, and the numbers of these pulses are shown in the top row.



different  $r_0$  values. This is illustrated in Figure 6, which is explained forthwith.

Let us start by considering the beat rhythm, which is shown by the bottom half-diamond. The user has control of the beat rhythm's  $r_0$  value, and the precise locations of some  $r_0$  values are indicated by the numbers directly above this diamond. As the value of  $r_0$  is changed, visualize a vertical line passing over the entire height of the figure; any horizontal diamond the line passes through is a higher-level well-formed rhythm that nests all lower-level rhythms (including the beat rhythm) at that  $r_0$  value. The horizontal extent of every such well-formed rhythm indicates the range of  $r_0$  values over which its own  $r_i$  values are between 1 and  $\infty$  (some  $r_i$  values for each well-formed rhythm are shown directly above its diamond). The thickness of each diamond gives a very approximate indication of the rhythm's evenness, which is maximal when  $r_i = 1$  and minimal when  $r_i \rightarrow \infty$  (as was illustrated in Figure 5). The numbers of long and short events in each well-formed rhythm are indicated by the figures inside each diamond. For example, the label 2:3 means there are two long and three short events. Note how each well-formed pattern swaps its numbers of long and short events as it passes through  $r_i = 1/1$ , as implied in Equation 2. Also

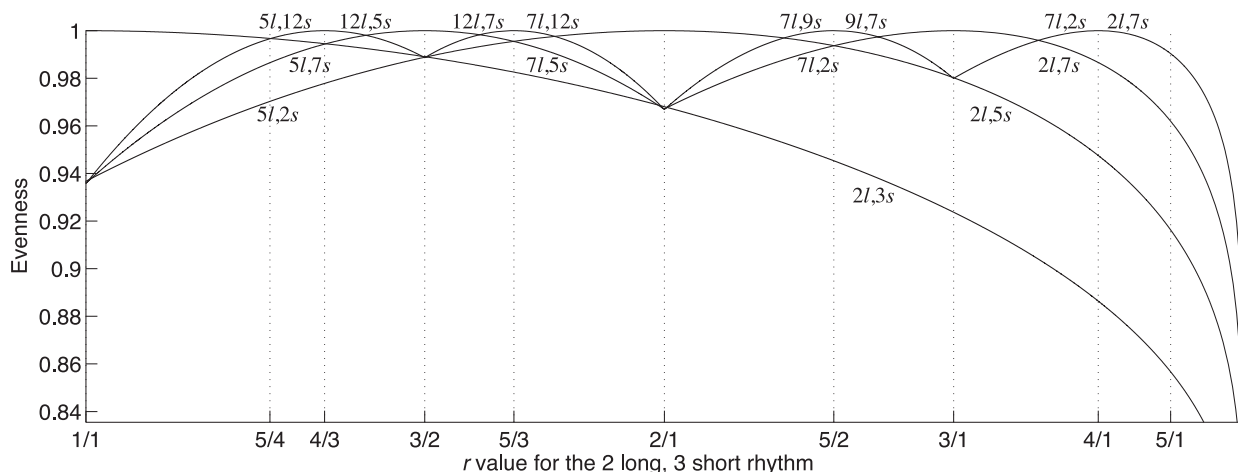
note that when  $r_i \rightarrow \infty$ , the well-formed rhythm is degenerate and its total number of events reduces because its short events vanish.

The level  $i$  of any well-formed rhythm is given by the number of diamonds the  $r_0$  line passes through to get to it. Hence the diamond labeled 5:2 and 2:5 is Level 1; the diamond labeled 5:7 and 7:5 and the diamond labeled 7:2 and 2:7 are both Level 2; the diamond labeled 7:9 and 9:7 and the diamond labeled 9:2 and 2:9 are Level 3; and so forth (each successive level is successively colored with alternating black and white). At any given level, there will always be a well-formed rhythm available (the apparent gaps at levels 3, 4, and 5 are actually filled by higher-level well-formed rhythms "above" the current figure).

Let us demonstrate how this works by considering the nested rhythms that are available when  $r_0 = 3/2$ . At this  $r_0$  value, the Level 0 beat rhythm—represented by the bottom diamond—is a rhythmic analogue of the standard pentatonic scale (which may help the visualization and auralization). It is analogous because the pentatonic scale has two large and three small steps and, assuming a standard twelve-tone equal temperament (12TET) tuning, a step-size ratio of 3:2 (its large steps are three semitones and its small steps are two semitones). The Level 1 rhythm in the next row up has seven

Figure 7. The evenness, over all  $r_0$  values, of the  $(2\ell, 3s)$  “pentatonic” rhythm and its next three higher “chromatic” levels. Note that the  $r_0$  values on

the horizontal axis refer to the step-size ratio for the lowest  $(2\ell, 3s)$  level, not that of the higher-level rhythmic streams.



nonisochronous pulses and nests the five-beat pattern. It comprises five long and two short pulses, and it has a step-size ratio ( $r$  value) of 2:1 (and so is a rhythmic analog of the 12TET diatonic scale with five two-semitone steps and two single-semitone steps). There is also a Level 2 rhythm that nests both the previous two levels (shown by the next black diamond up), which contains twelve pulses. When  $r_0 = 3/2$  for the Level 0 rhythm,  $r_2 = 1$  for the Level 2 rhythm. This means that all the latter's pulses are isochronous, hence this is the rhythmic analog of the equal-tempered chromatic scale. All higher-level rhythms have twelve isochronous pulses.

As the value of the beat rhythm's  $r_0$ -slider is changed, MeanTimes automatically selects the appropriate well-formed rhythm for each higher level, and finds the appropriate event timings by recursion of Equations 2–4.

### Evenness of the Well-Formed Hierarchy

We previously showed, in Figure 5, how the evenness of each possible five-beat well-formed rhythm changes over all  $r_0$  values. Clearly, the same process also occurs for the higher-level well-formed rhythms too. In general, each will have an analogous curve (maximal at  $r_i = 1$ , minimal at  $r_i \rightarrow \infty$ ) but over a smaller  $r_0$  range as levels are ascended. An example, for the well-formed rhythms, up to Level 3, that nest

the two-long-, three-short-beat rhythm is illustrated in Figure 7.

In MeanTimes's interface, the evenness values for the beat-level and each of the three higher-level pulses are displayed as bar graphs, which are updated in real time as the  $r$  slider is moved. The color of the bar also indicates whether each rhythm level is coherent (its  $r$  value is less than two), and each level's  $r$  value is also displayed.

### Extensions and Modifications of Well-Formed Rhythms

In the earlier section “Well-Formed Words,” we formalized the splitting of each long beat into a new long and short beat with the mapping  $\ell \mapsto \ell s$ . We could have chosen the opposite ordering:  $\ell \mapsto s\ell$ . This implies changing the first morphism  $\ell \mapsto \ell s$  and  $s \mapsto \ell$  (as defined earlier) to  $\ell \mapsto s\ell$  and  $s \mapsto \ell$ ; and changing the second morphism  $\ell \mapsto \ell s$  and  $s \mapsto s$  (as defined earlier) to  $\ell \mapsto s\ell$  and  $s \mapsto s$ . For convenience, we will refer to levels resulting from the original morphisms as *clockwise*, and those originating from the just introduced alternatives as *counterclockwise*. Interestingly, clockwise and counterclockwise levels are identical up to rotation (or reflection). This means that by independently changing the directionality of the levels in a hierarchy, their relative rotations change.

This results in differing beats being duplicated across the levels. Using a pitch-based analogy, consider the pentatonic scale A–C–D–E–G. The clockwise morphism gives A–B–C–D–E–F♯–G; the counterclockwise morphism gives A–B♭–C–D–E–F–G. Both resulting scales are diatonic, but the scale degrees of the notes duplicated in the lower-level pentatonic scale differ. Numbering from the first degree of the major scale (G in the first diatonic scale, and F in the second), the first scale is duplicated at  $\hat{1}$ – $\hat{2}$ – $\hat{4}$ – $\hat{5}$ – $\hat{6}$ , while the second scale is duplicated at  $\hat{2}$ – $\hat{3}$ – $\hat{5}$ – $\hat{6}$ – $\hat{7}$ . These two versions of the diatonic hierarchy are illustrated, respectively, in Figures 4a and 4b. Each rhythmic level in MeanTimes has a directional toggle allowing a wide variety of different patterns of duplications and rotational relationships between levels.

Another interesting modification that can be applied in MeanTimes is to use what we call the *complement* of any given rhythmic level. This can be used to eliminate beat duplication across levels. As just described, the next level above the pentatonic scale C–D–E–G–A is the diatonic C–D–E–F♯–G–A–B (assuming a clockwise morphism). Instead of using all the notes in the higher level, it is possible to use just those notes that are not found in the lower level; in this case, F♯ and B. In this way, each higher level only splits long beats of the lower level (it fills in the gaps), it no longer duplicates the lower level. This can be used to produce multiple interlocking patterns that, in combination, fully delineate a pulse. Interestingly (as can be shown from Amiot 2007, Proposition 3.2), these complementary patterns are also well-formed. This is illustrated in Figure 4, where the patterns emanating from a rhythm analogous to the pitch-class set C–F–G in 12TET (Level 0 has two large and one small beat, and  $r_0 = 5/2$ ) are shown when complementary mode is not engaged for any level (Figures 4a and 4b), and when it is engaged for all levels (Figure 4c).

With the hierarchy of well-formed rhythms in place, it is easy to introduce the possibility of “chromatic” alteration, or modification, of any well-formed rhythm. We use the term *chromatic* to re-emphasize the relation between meter and scale, because a chromatic modification can create a new

beat size, distinct from  $s$  or  $\ell$ , rather as a chromatic modification of a scale member (even if it is to an equal-tempered alternative pitch) may create new interval step sizes in the result. To do this, we simply displace any given event by an amount that is equivalent to the new pulse size that is introduced by the next higher-level rhythm.

As shown in Equation 3, when a Level 0 well-formed pattern is advanced to Level 1, the long beat  $\ell_0$  is split into two sizes  $\ell_1$  and  $s_1$ . Furthermore, one of these takes the size  $\ell_0 - s_0$ , and the other takes the size  $s_0$ . By analogy with musical scales, we take the new size  $\ell_0 - s_0$  as the size of the chromatic alteration; that is, a Level 0 event is “sharpened” by delaying it by  $\ell_0 - s_0$ , and it is “flattened” by advancing it by  $\ell_0 - s_0$ . In MeanTimes, a specific alteration is achieved by dragging a polygon vertex, and it will snap to a pulse position from the next higher level. The corresponding beat at all lower levels will also move in tandem. Making any single such modification results in a number of different possibilities for the form of the resulting rhythm: The new rhythm may simply be a transposition (mode change) of the original; it may create a rhythm with the same two beat lengths but arranged in a non-maximally even form (not making a Christoffel word, or rotation thereof); it may create a rhythm with three beat lengths; or it may create a rhythm with four beat lengths.

We now give examples of such possibilities by looking at analogous chromatic alterations of the C major scale. If just the F is sharpened, the resulting scale is simply a transposition of the original (it has moved from C major to G major or, equivalently, from Ionian to Lydian mode). If just the E is flattened, the resulting scale is the ascending melodic minor, which contains the original two step sizes (major and minor seconds) but now arranged in a non-well-formed pattern. If just the G is sharpened, the resulting scale is the non-well-formed harmonic minor scale, which now contains three step sizes (there is an additional augmented second between F and G♯). Similar results are obtained if just the A is flattened, which gives the harmonic major scale.

As shown in Figure 8, all but the first of these modifications will reduce the analogous rhythm’s evenness. This is of obvious compositional utility as

Figure 8. The evenness, over all  $r$  values, of five different seven-beat patterns. From most to least even: diatonic (e.g., C-D-E-F-G-A-B),

melodic minor (e.g., C-D-E $\flat$ -F-G-A-B), harmonic major or minor (e.g., C-D-E-F-G-A $\flat$ -B or C-D-E $\flat$ -F-G-A $\flat$ -B), and double harmonic (e.g.,

C-D $\flat$ -E-F-G-A $\flat$ -B). The harmonic major and harmonic minor are inversionally equivalent and so have identical evenness.

Figure 9. The entropies of different seven-beat patterns. Entropies are calculated for sequences ( $n$ -tuples) of consecutive event sizes.

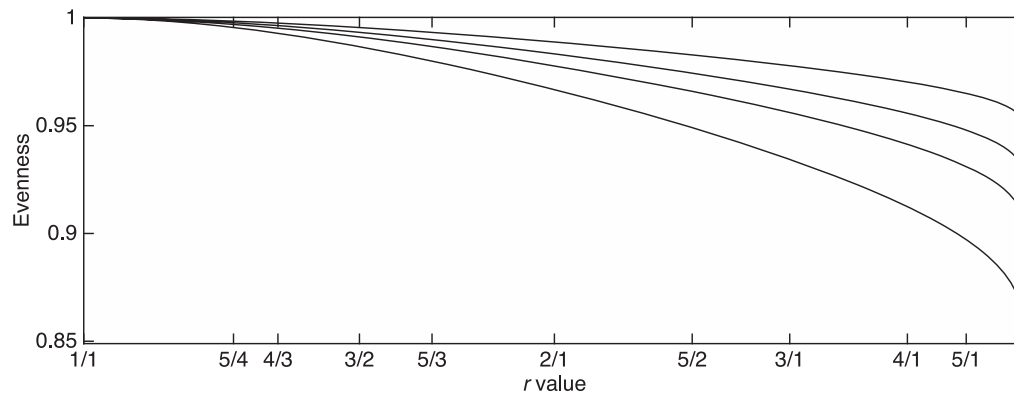


Figure 8

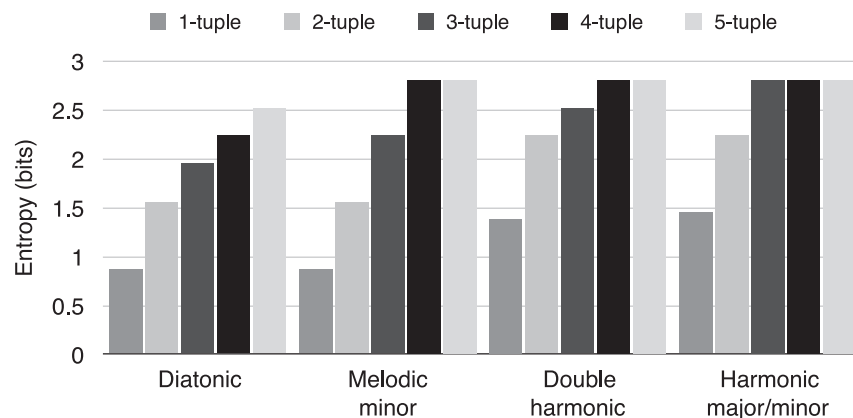


Figure 9

a motivic transformation device, but also of interest in terms of the resultant change in metricality and rhythmic transparency.

The  $r$  value has no effect on the entropy of event sizes, but the actual form of the scale does. This is shown in Figure 9, which shows the effect on entropy of the given modifications of well-formedness. The entropies are calculated from probability distributions over  $n$ -tuples of consecutive events, with  $n$  taking values from one to five. We will show how these are calculated by considering the diatonic pattern of  $\ell ls \ell ls$ . Stepping through the diatonic pattern from left to right, the following 2-tuples occur:  $\ell \ell$ , then  $\ell s$ , then  $s \ell$ , then  $\ell \ell$ , then

$\ell \ell$ , then  $\ell s$ , and finally  $s \ell$  (proceeding from the last element to the first). Hence, the probability mass function over all possible 2-tuples is  $p(\ell \ell) = 3/7$ ,  $p(\ell s) = 2/7$ ,  $p(s \ell) = 2/7$ , and  $p(ss) = 0$ . The entropy (in bits) of a probability mass function is calculated by  $H(p) = -\sum_i p_i \log_2 p_i$ ; hence the entropy for this example is 1.56.

Stepping through the diatonic pattern from left to right, the following 3-tuples occur:  $\ell ls$ , then  $\ell s \ell$ , then  $s \ell \ell$ , then  $\ell \ell \ell$ , then  $\ell \ell s$ , then  $\ell s \ell$ , and finally  $s \ell \ell$ . Hence, the probability mass function over 3-tuples is  $p(\ell \ell \ell) = 1/7$ ,  $p(\ell \ell s) = 2/7$ ,  $p(\ell s \ell) = 2/7$ ,  $p(s \ell \ell) = 2/7$ ,  $p(\ell s s) = 0$ ,  $p(s \ell s) = 0$ ,  $p(s s \ell) = 0$ , and  $p(s s s) = 0$ . This has an entropy of 1.95. And so forth.

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## Compositional, Analytical, and Perceptual Implications

MeanTimes permits systematic generation and perturbation of well-formedness of a meter, and continuous variation of long/short beat ratio  $r$ , which produces continuous variation in the evenness of the rhythms generated. Furthermore, different combinations of the levels of the metrical hierarchy (where increasing level corresponds to shortening event lengths) may produce different pulse and apparent beat positions, in addition to those of the lowest beat level, such that new meters are created without change in period length. Compositionally, all these features are of interest for systematic control, and many of the rhythmic instantiations would not be readily accessible by manual composition nor manually performable by instrumentalists. They are thus quintessentially computer-dependent rhythmic compositional devices, permitting rhythmic morphing that has as yet been little explored.

To make this conclusion clear, we can consider some of the compositional rhythmic complexities that have been used in instrumental works and then, in contrast, some mechanical or computerized alternatives. Serialization of rhythm or the use of irrational rhythms results in rhythmic and metrical irregularity (even if, particularly in the case of serialization, rhythmic patterns may repeat often). But these techniques do not readily allow continuous variation of event-size ratios, especially in the case of serialized rhythm. With irrational rhythms or spatial (i.e., proportional) notation any sequence of event sizes and durations can be notated, but the approach does not in itself imply a systematic way of controlling such sequences, nor would the resultant score be likely to receive a performance that completely fulfilled its specifications.

Elliott Carter's metric modulation (Hobert 2010), or its improvised relatives in some work of Miles Davis (1964–1966) or subsequent groups such as Chick Corea's ensemble Circle (Dean 1992), offer devices for abrupt metrical change, with or without change in period length. For example, Carter often notates transitions in which a unit of length (be it pulse or bar) is retained but subdivided by a different integer. Related devices occur in Davis's work, but

improvisation does allow more continuous variation in such changes, as heard in recordings such as those by Circle and others.

It is perhaps in the mechanically performed and rhythmically complex music of Conlon Nancarrow that we find an antecedent to MeanTimes, although (as far as we are aware) there is little in his writing or in analyses of his work to suggest a systematic use of the ideas developed here. Nancarrow was most interested in coexistence of multiple tempi and meters, whose barlines rarely coincide, and in trying to "synthesize the two opposing conceptions [divisive, additive] of rhythm" (Gann 2006, p. 7). There is no doubt that a player-piano performance system such as he used could, like a present-day computer, realize the rhythmic patterns MeanTimes can produce.

The combination of perturbing well-formedness and morphing rhythmic ratios and periods means that MeanTimes provides a novel compositional and improvisational tool, and further extensions to the algorithm can be readily envisaged. Using the algorithmic approach described here, intelligent compositional input is still required: not all well-formed rhythms, or transitions between them, will sound appropriate. Furthermore, effective choices for the timbres or pitches used for the multiple levels are still required. Our composing and performing experience to date with MeanTimes suggests, however, that we can indeed make interesting music with it, including morphing in and out of a feeling of groove, and finding some deeply nonisochronous grooves in the process. We hope to use it more extensively in composition and improvisation in the future. An audio recording of the first performance with it, given by the two authors at the Sydney Conservatorium of Music in 2014, is provided as an online supplement to this article ([www.mitpressjournals.org/doi/suppl/10.1162/COMJ\\_a.00343](http://www.mitpressjournals.org/doi/suppl/10.1162/COMJ_a.00343)). We also include a variety of demonstrations.

Composition and improvisation are both creative and analytic processes, and MeanTimes has the potential to contribute also to the analysis of rhythm and its performance. At the broadest level, the theories of well-formedness and evenness offer new descriptive and comparative approaches to



rhythmic and metrical structure. MeanTimes may also offer an enhanced approach to the analysis of rhythmic variation in performance. It is not most commonly the case that metrically conceived Western or Asian classical music is performed with a precision approaching isochrony or precise repetition of beat sizes. Rather, variation in such beat sizes is used as an expressive vehicle. By considering the extrema of beat-size ratios exhibited in timing variations, in relation to the evenness constraints of MeanTimes, further insight may be obtained as to how they may operate and be optimized for expression.

A composer's or improviser's subjective impressions of the interest or degree of groove of rhythmic patterns are of course of central importance to that person, even though they may be capable of in-depth rhythmic analysis of the kind just indicated. But what of the average (musically untrained) listener? Clearly subjective impressions (i.e., perception) are necessarily dominant, as they normally do not have any analytic framework to use as an alternative. The study of perception of metricality, particularly by nonmusicians, can be based on at least two levels of response. First, does a person reproducibly identify a cycle length as the meter is enunciated? This can be tested in tapping experiments, by asking a participant to tap along with the repeating metrical pattern such as to represent any repeating features the listener notices. The responses can be quantified in terms of the proportion of times the cycle length is correctly tapped. In addition, if a respondent taps in several different positions within the pattern, these data begin to address the second level of analysis of perception of metricality: the question of whether listeners can also identify the repeating subcycles, or beats, of the meter.

In the absence of strong perceptual data, it is not justified to speculate extensively about predictors of metricality. But, overall, we predict that well-formed meters will be more readily perceptible than others and that the more even a meter, or its rhythmic instantiation, the more well perceived that meter will be. These predictions are based both on the overall expectation that the higher the entropy of the pattern, the less likely it is that the period will be detected and on our own

experience in performing with and evaluating the rhythmic outputs of MeanTimes.

## Conclusion

Our mathematically driven approach to meter treats events on pulse and beat positions with equity in relation to evenness, such that well-formed patterns include not just the simple regular isochronous beats of conventional  $\frac{3}{4}$ ,  $\frac{4}{4}$ ,  $\frac{6}{8}$ , and so forth, but also patterns comprising pulses organized in two beat sizes—such as those found in Balkan music. Furthermore, well-formedness can in this way be defined for patterns with no underlying isochronous pulse (the beat or pulse lengths have no common divisor). This means their long and short beat sizes can be covaried across a continuum, thereby creating rhythms and meters that narrowly miss perfect isochronies, or more radical patterns where obvious isochronies are absent. Such variation also encompasses a variety of common non-well-formed meters (again with no requirement for isochrony at the pulse level). In these ways, well-formed patterns can be used as the starting point for creating systematic families of rhythms that may have musical and perceptual interest.

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